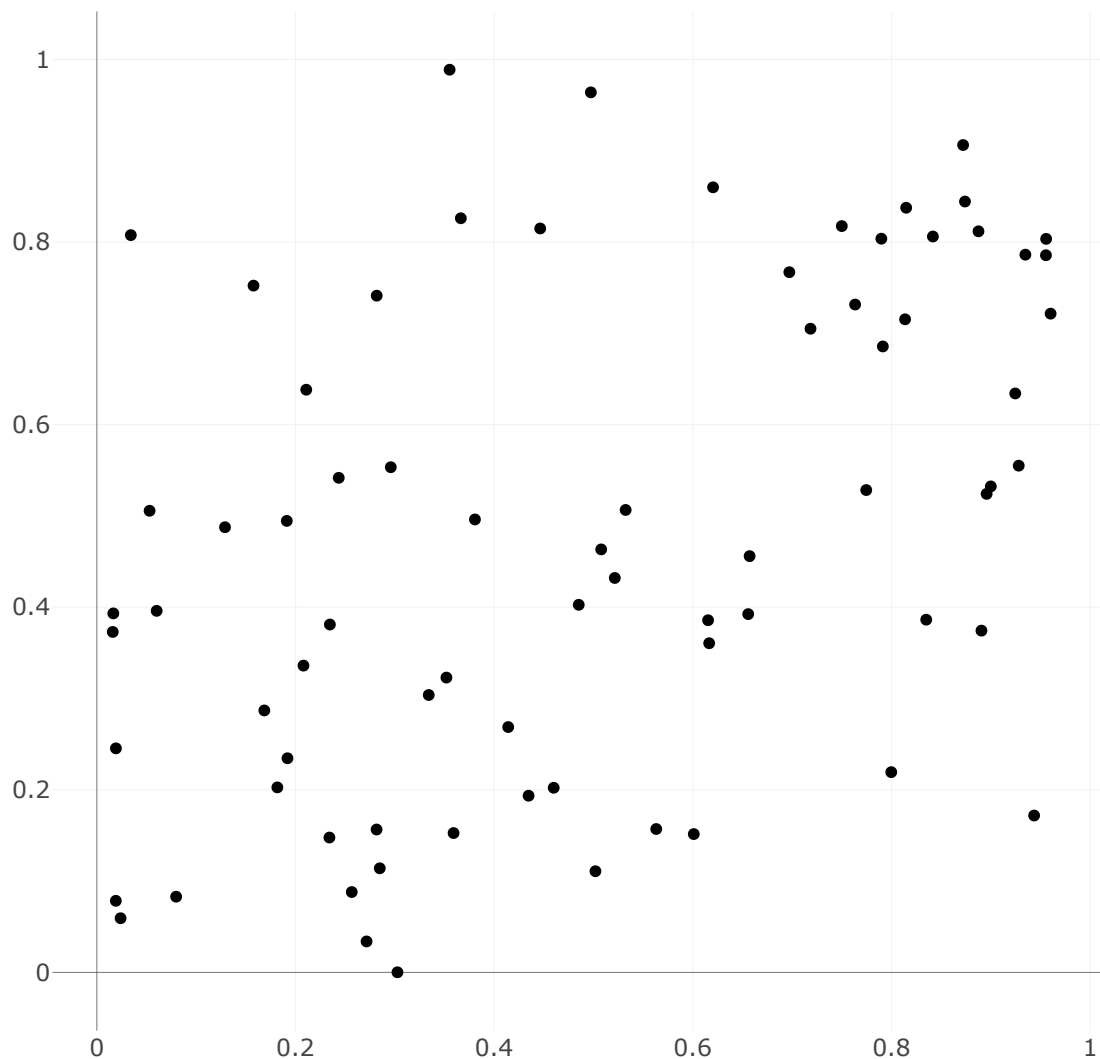


Facility Location Model

Data: demand points

The location of the demand points are randomly generated and drawn from the uniform distribution.

Demand points	75
Max distance customer $\rightarrow$ facility	0.250

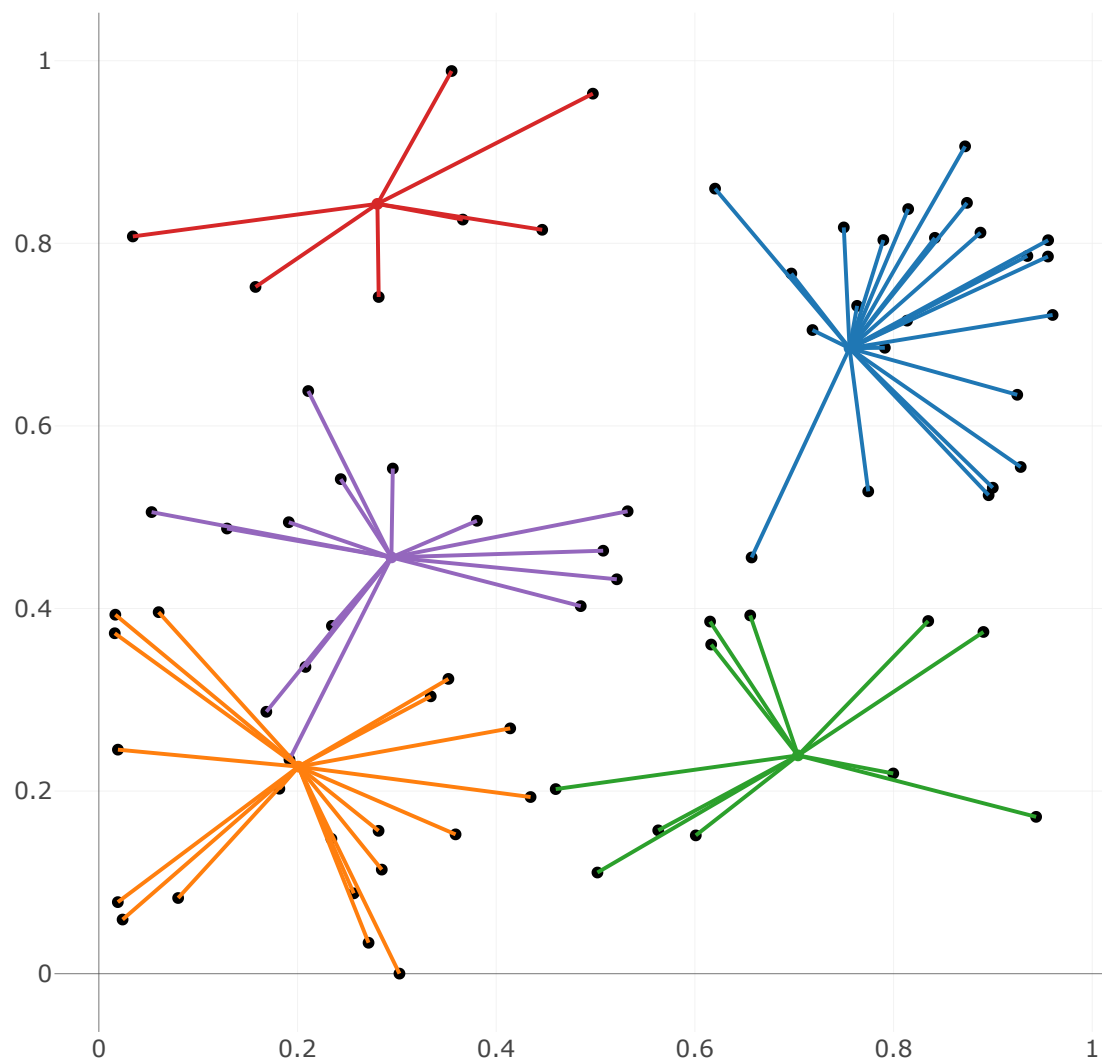


Model 1: results

Model 1 finds the minimum number of facilities needed to serve all customers and obey the maximum distance constraint. To make the model quadratic, the constraint is formulated as a maximum quadratic distance constraint.

It is noted that the solution is **not** an assignment with shortest distances between customers and facilities.

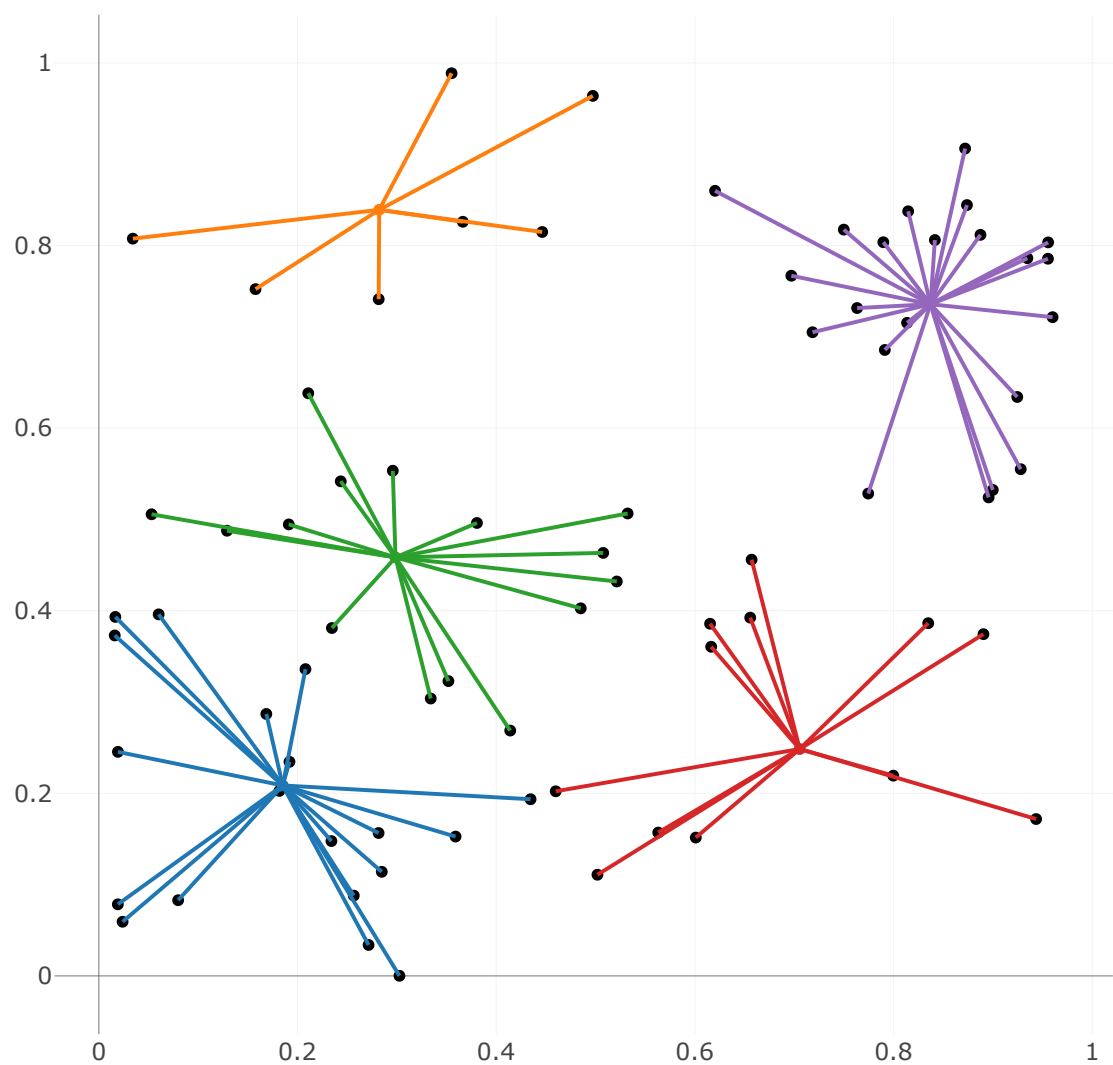
Number facilities needed (min)	5
Sum distances	13.095
Sum squared distances	2.558
Max distance	0.250
Binary variables	760
Solver time	8.750
Nodes	14947



Model 2: results

Model 2 find the best location of the facilities and the optimal assignment of customers to the facilities. It uses the number of facilities found in model 1.

Sum squared distances (min)	2.181
Sum distances	11.902
Max distance	0.250
Binary variables	375
Solver time	1.890
Nodes	6950

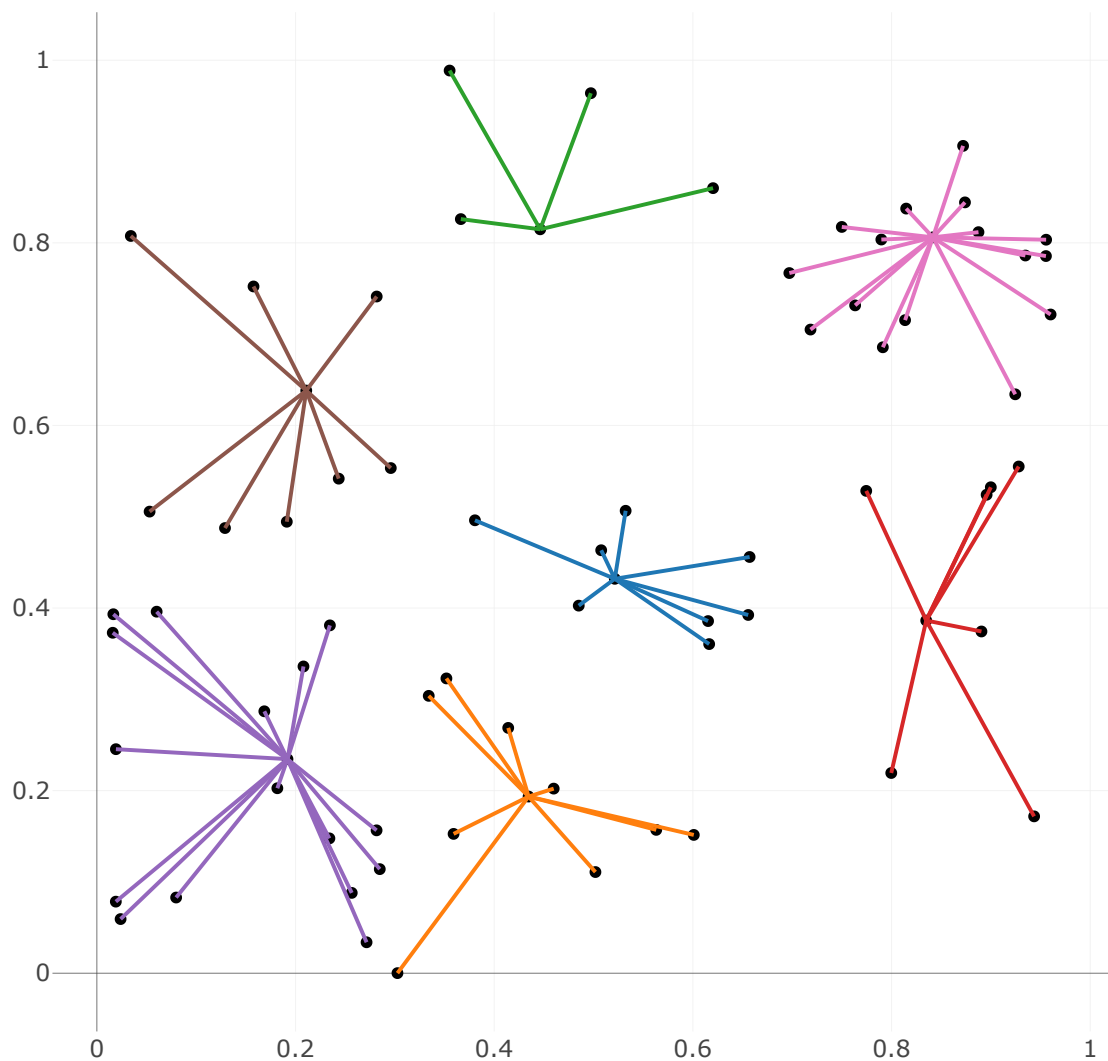


Model 4: results

Model 4 uses the demand points as candidate locations for the facilities. This is an easy MIP. It is solved here in two stages: first find the optimal number of facilities and then find the best locations.

Number of facilities	7
Sum squared distances	1.473

Sum distances (min)	9.217
Max distance	0.245
Binary variables (min numFacs)	1152
Solver time (min numFacs)	0.047
Nodes (min numFacs)	0
Binary variables (min totDist)	1152
Solver time (min totDist)	0.062
Nodes (min totDist)	0



Model 5: results

Model 5 is tracing the trade-off between `maxdist` (the maximum distance limit) and `numfacs` (the number of facilities). The results are based on Medoid based model: the candidate locations of the facilities is the set of demand points.

numFacs	maxDist
1.000	0.622
2.000	0.521
3.000	0.375
4.000	0.337
5.000	0.291
6.000	0.260
7.000	0.245
8.000	0.210
9.000	0.208
10.000	0.190

